American Computer Science League

2025-2026 ● Contest 1: Short Problems ● Intermediate Division

1. Computer Number Systems

Evaluate the following and express the answer in octal.

$2025_8 - 202_8 + 20_8 / 2_8$

A. 1721
B. 1367
C. 1633
D. 1623
E. 1523

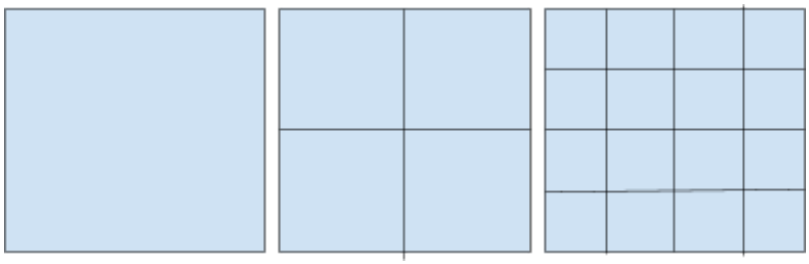
2. Computer Number Systems

The next number of the Fibonacci sequence is formed by adding the previous two numbers. The numbers are 0, 1, 1, 2, 3, 5, … How many times does 1 appear in the binary representation of the first 10 Fibonacci numbers starting at 0?

A. 4
B. 10
C. 16
D. 20
E. 22

3. Recursive Functions

Given a 16 cm by 16 cm square.
Repeatedly divide the square into 4 equally sized squares by connecting the midpoints of the sides vertically and horizontally.



Stop after subdividing the square 5 times.
What is the perimeter of the smallest square?

A. 0.5
B. 1
C. 2
D. 4
E. 8

4. Recursive Functions

Find $f(16, 12)$ given:

$$f(x,y) = \begin{cases} f(x-2, y+1) + y & \text{if } x > y \\ f(x+1, y) - x & \text{if } x < y \\ \dfrac{x}{2} & \text{if } x = y \end{cases}$$

A. 6
B. 7
C. 13
D. 14
E. 18

5. What Does This Program Do?

Which of the following input values of a, b, and c will produce an output of 7, 24, 25?

```
input a, b, c
if c == 0 then
     c = sqrt(a^2 + b^2)
else
    if b == 0 then
        b = sqrt(c^2 - a^2)
     else
        a = sqrt(c^2 - b^2)
     end if
end if
output a, b, c
```

A. 0, 24, 25
B. 7, 0, 25
C. 7, 24, 0
D. All of the above
E. None of the above

6. What Does This Program Do?

What is output by the following program?

```
a = 13
b = 21
n = 8
while sqrt(b) != int(sqrt(b))
     n = n + 1
     c = a + b
     a = b
     b = c
end while
output n
```

A. 12
B. 10
C. 11
D. 16
E. 13

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2025-2026 ● Contest 1: Short Problems ● Intermediate Division

1. 计算机数字系统

求以下表达式**的值**，并以八进制表示结果。

$2025_8 - 202_8 + 20_8 / 2_8$

A. 1721
B. 1367
C. 1633
D. 1623
E. 1523

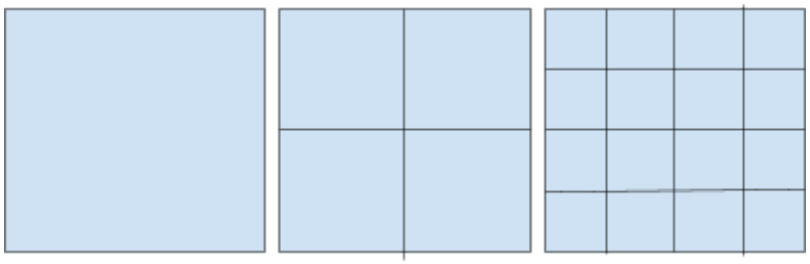
2. 计算机数字系统

斐波那契数列的下一项由前两项相加得到。**现有**数列：0、1、1、2、3、5、…… 从 0 开始，前 10 个斐波那契数在二进制表示中“ 1 ”出现了多少次？

A. 4
B. 10
C. 16
D. 20
E. 22

3. 递归函数

已知一个**边长**为 16 cm 的正方形。
通过在水平方向和垂直方向连接各边的中点，将该正方形反复划分为 4 个等大的正方形。



在共细分 5 次后停止。
最小正方形的周长是多少？

A. 0.5
B. 1
C. 2
D. 4
E. 8

4. 递归函数

根据**下列表达式，**求$f(16,12)$ ：

$$f(x,y)=\begin{cases} f(x-2,y+1)+y & if\ x>y \\ f(x+1,y)-x & if\ x<y \\ \dfrac{x}{2} & if\ x=y \end{cases}$$

A. 6
B. 7
C. 13
D. 14
E. 18

5. 这个程序是做什么的？

下列哪组 a、b 和 c 的输入值会输出 7、24、25？

```
input a, b, c
if c == 0 then
     c = sqrt(a^2 + b^2)
else
    if b == 0 then
        b = sqrt(c^2 - a^2)
    else
        a = sqrt(c^2 - b^2)
    end if
end if
output a, b, c
```

A. 0, 24, 25
B. 7, 0, 25
C. 7, 24, 0
D. **以上所有都能**
E. **以上所有都不能**

6. 这个程序是做什么的？

以下程序的输出是什么？

```
a = 13
b = 21
n = 8
while sqrt(b) != int(sqrt(b))
     n = n + 1
     c = a + b
     a = b
     b = c
end while
output n
```

A. 12
B. 10
C. 11
D. 16
E. 13